# Baseline AI Test Report (Week 6)

COMP 441 Software Analysis and Testing - Group Project "AI-Augmented Test Suite"

| Field | Detail |
|---|---|
| Track | C (Healthcare / Patient-Records Data Validation) |
| Repository | keitsengSIR/trackc_group_assignment (Django 4.2 hospital management system) |
| Author / role | Maano Dikgang, AI Test Engineer |
| Date | 15 October 2026 |
| Branch | maano_branch |

## 1. Tools and versions

| Item | Version |
|---|---|
| Python | 3.12.10 |
| Django | 4.2.9 |
| pytest / pytest-django / pytest-cov | 9.1.1 / 4.14.0 / 7.1.0 |
| coverage.py | 7.16.2 |
| AI tool used for test generation | Claude (Anthropic), claude.ai chat interface |
| Model shown in the UI | Claude Sonnet 5.5 |
| Tier | Free |

pytest is configured in `pytest.ini` with `--nomigrations`, because the repository's migration chain fails on a fresh database: `website.0003_auto_20250625_2353` tries to create `website_contactinquiry`, which already exists. Tests build their tables directly from the models instead.

## 2. Scope and justification

Tests were generated for three targets only:

- `PatientRegistrationForm` and `PatientSearchForm` (`patient/forms/patients.py`)
- The `Patient` model: `age`, `full_name` and `patient_id` generation in `save()` (`patient/models.py`)
- `extract_common_keywords` (`patient/services.py`)

Reasons: these files had 0% coverage before any AI work, they are small enough for mutation testing in Week 7, and they carry the Track C risks (patient data fields, dates of birth and identifiers). Views were excluded because they need authentication, sessions and role middleware, and add about 1,500 untested statements.

## 3. Prompt log

**Prompt #1** (full text, including the code under test, saved in `ai-baseline/prompt1_full.txt`):

- Role: write pytest-django unit tests for the scoped targets.
- Rules: do not modify source; single output file; use django_db only where needed; synthetic data only; cover valid, invalid, boundary and missing-field cases; every test must assert specific values

or errors; return only code.

- Follow-up messages sent: none.

The raw output was saved unedited to `tests/ai_generated/test_patient_ai.py` and committed as `2e886b6` before any change was made. No edits were needed to run it.

# 4. Coverage results

## Before AI (existing suite, 6 tests)

Run with `--cov=patient --cov=appointment --cov-branch`: 2,845 statements, 2,451 missed, 734 branches, **11% total**. Evidence: `ai-baseline/baseline_before_ai.txt`.

## AI-only suite (118 tests collected: 117 passed, 1 xfailed, 0 failed)

Run with `--cov=patient --cov-branch`. Evidence: `ai-baseline/baseline_ai_only.txt`.

| File | Stmts | Missed | Branches | Partial | Cover | Before AI |
|---|---|---|---|---|---|---|
| patient/forms/patients.py | 24 | 0 | 4 | 1 | 96% | 0% |
| patient/services.py | 32 | 0 | 2 | 0 | 100% | 0% |
| patient/models.py (whole file) | 323 | 81 | 36 | 0 | 68% | 65% |
| Total (patient only) | 1,633 | 1,200 | 384 | 1 | 22% | not comparable |

- The two totals use different coverage scopes, so compare per file, not totals.
- models.py contains many classes outside the scope. Inside the Patient class (lines 10 to 91) the only line not executed is line 78, the return statement of get_absolute_url; all other missed lines (133 onward) belong to other classes.
- The partial branch in patients.py (49->52) is the if in save() that maps medical_conditions to chronic_diseases; its false path is never taken.
- In services.py most statements are imports, so 100% overstates the logic covered.

# 5. Assertion-quality assessment

**Existing tests (before AI).** test_patient_validation.py defines validate_patient_age inside the test file, so it tests only itself and no application code. test_integration.py ends with assert True, so it cannot fail. Neither provides real evidence of testing.

**AI-generated tests, strengths.** Exact error messages are asserted; max_length is tested on both sides of each limit; Patient.age is tested around birthdays, leap day and the year boundary; extract_common_keywords is tested for case, stopwords, the length-3 cutoff and the limit parameter.

**AI-generated tests, weaknesses.**

1. **Tests that assert buggy behaviour as correct.** The AI described what the code does, not what it should do. These pass today and would fail if the bug were fixed:

- test_registration_form_accepts_future_date_of_birth (year 2999 accepted)
- test_patient_age_future_date_of_birth_is_negative (age is -1)
- test_registration_form_save_overwrites_chronic_diseases_with_medical_conditions (an empty medical_conditions erases chronic_diseases)
- test_patient_id_collision_after_deletion_raises_integrity_error (IDs come from count() + 1)

2. **A genuine failing check recorded as xfail.**
test_generated_patient_id_fits_within_field_max_length is XFAIL: the generated ID P-2026-0001 is 11 characters but patient_id has max_length=10. SQLite does not enforce this, so it passes silently in the database.

3. **Missing business rules.** The 0 to 120 age rule in the existing placeholder test is not enforced anywhere in the form or model, and phone fields accept any text up to 15 characters. The AI tests do not probe either gap.

## 6. Limitations observed

- High coverage (96% to 100% on the scoped files) does not show that the tests would catch a defect; mutation testing in Week 7 is needed to measure that.
- The AI documented existing defects instead of flagging them, so a green suite can protect bugs.
- The AI could only see the code, not the intended requirements, so requirement-driven cases (for example a plausible date-of-birth range) were absent.
- All 117 passing tests ran without edits; no hallucinated names were observed in this run.

## 7. Next steps (Week 7)

- Run mutation testing on the scoped code and compute the v1 score.
- Analyse at least 8 surviving mutants and write manual tests to kill them.
- Report the confirmed defects (ID length, chronic_diseases overwrite, no date-of-birth validation) to the team.

## 8. Evidence index

| File | Content |
|---|---|
| pytest.ini | pytest configuration |
| ai-baseline/baseline_before_ai.txt | coverage before AI |
| ai-baseline/baseline_ai_only.txt | coverage of the AI-only suite |
| ai-baseline/prompt1_full.txt | prompt #1 as sent |
| tests/ai_generated/test_patient_ai.py | raw AI output (commit 2e886b6) |